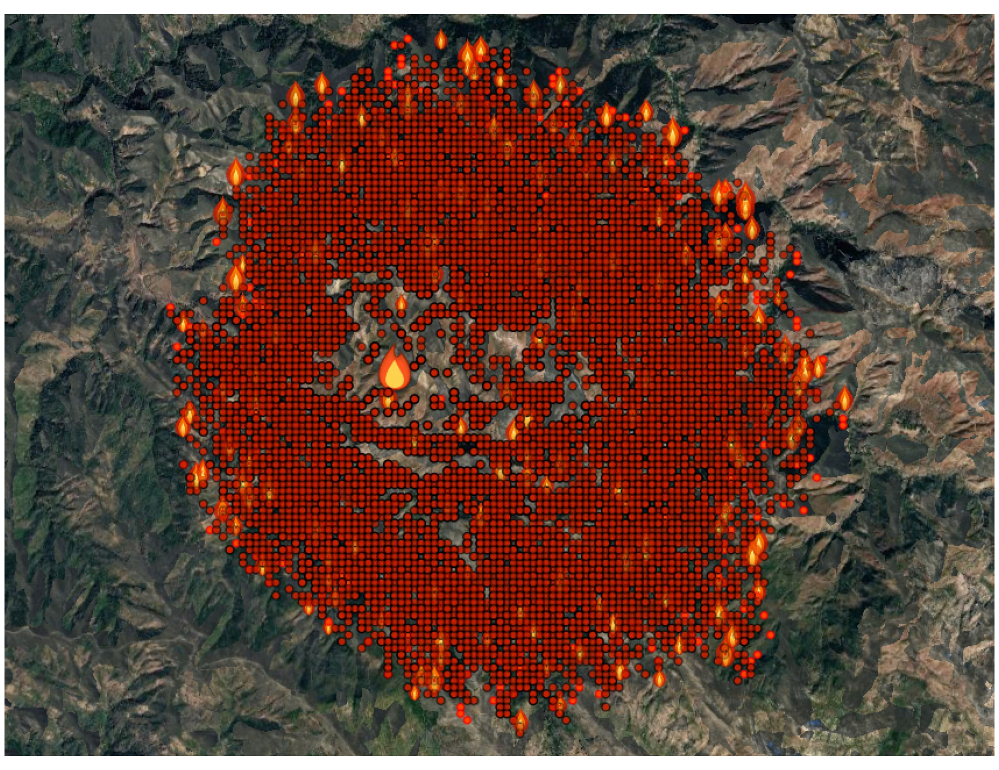
How the dog decides

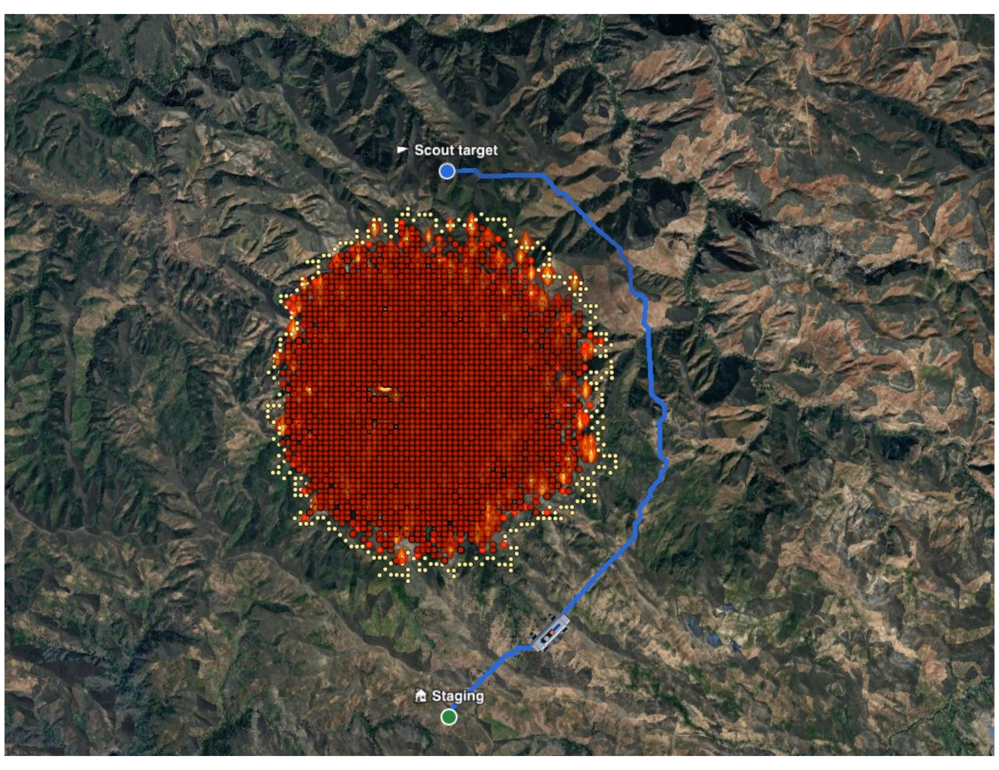
1 Predict



50 m cells **15 min** steps

Forecasts when each cell will burn.

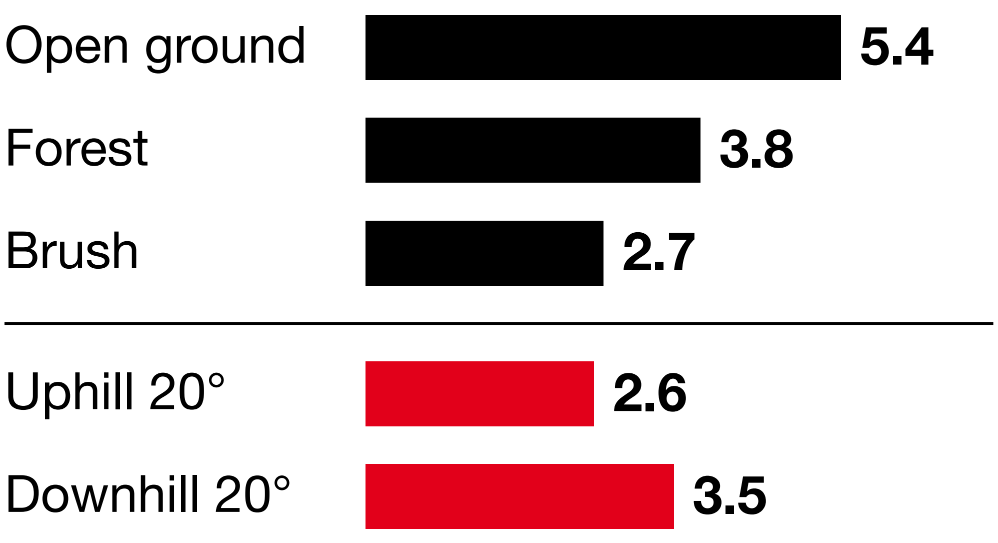
2 Safe route



Blue route to the scout target

Keeps 100 m from the forecast fire, with 30 min in hand.

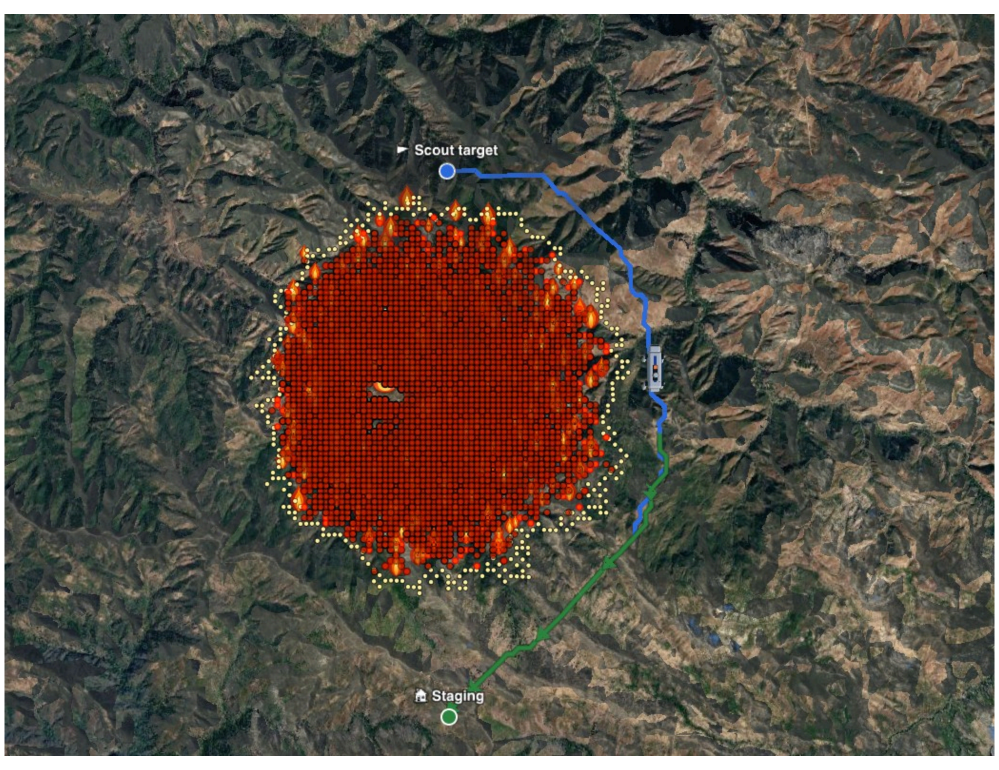
2b Route speed



Walking speed in **km/h**

The route is timed at these speeds. The dog never crosses water or slopes over 35°.

3 Escape path



Green escape path home

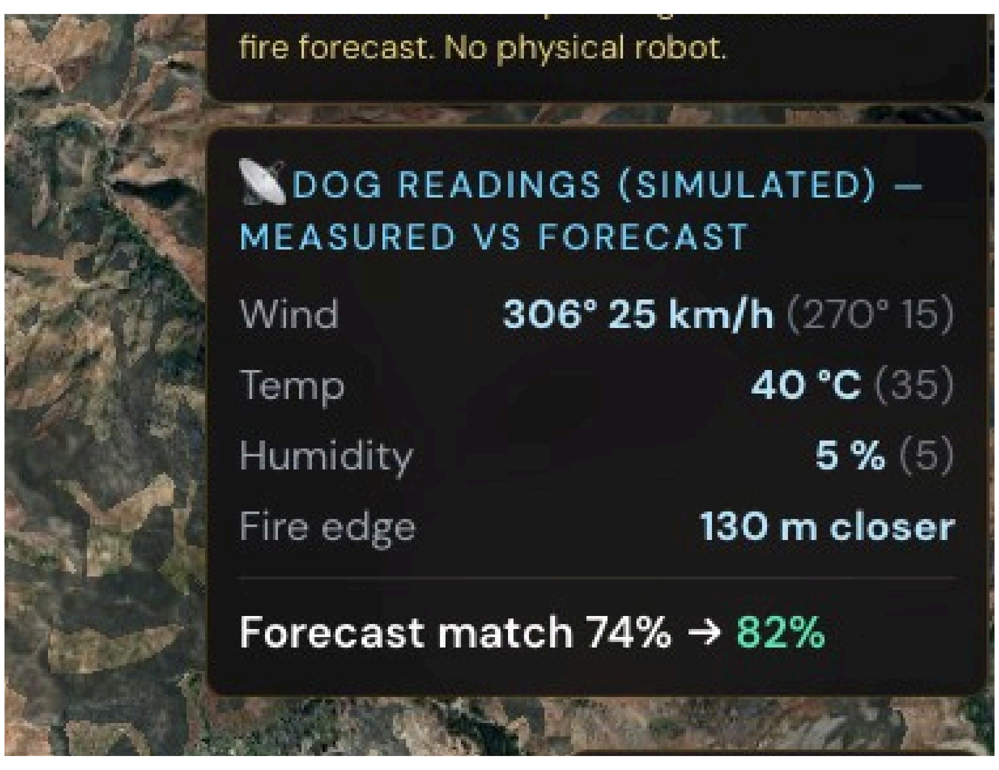
Aims for a way home that stays 100 m clear for an hour.

4 Scout

Wind, temperature, humidity, fire edge

Measures what the forecast only guessed.

5 Feedback



fire forecast. No physical robot.

DOG READINGS (SIMULATED) — MEASURED VS FORECAST

Wind	306° 25 km/h (270° 15)
Temp	40 °C (35)
Humidity	5 % (5)
Fire edge	130 m closer

Forecast match 74% → 82%

Launch **83%** Shift **74%** Report **82%**

The readings improve the next forecast.

Simulator views on real terrain. Yellow dotted edge: the forecast the dog plans on; the real fire is hidden from it until its sensors see it. Robot image: design concept; sensors are simulated.

Status and next

Done	Next
ML fire forecast trained and tested	Forecast + planner on the dog's controller
3D browser simulator on real terrain	Real sensors feeding the forecast
Containment planner (search)	Scout the planned containment line
Robot-dog loop in mock simulation	Tests on real robot dogs

Open-source code and demo video

Data and references

Code: github.com/sxz2012/wildfire-simulation

NASA FIRMS (VIIRS active-fire detections); Open-Meteo weather archive; ESA WorldCover 10 m land cover; Cesium World Terrain; Google Photorealistic 3D Tiles

Rothermel, R. C. (1972). *A mathematical model for predicting fire spread in wildland fuels.* USDA Forest Service INT-115.

Chen, T. & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. *KDD '16.*